



Figure 10: Dialog passwordbox and Xdialog inputbox widgets



Figure 11: Dialog timebox and Xdialog msgbox widgets

```
execute(dwdgtatrb["msg"].replace('message', msg))

else:
    msg = 'Incorrect password, better luck next time.'
    execute(dwdgtatrb["msg"].replace('message', msg))

if '__main__' == __name__:
    main()
```

Figures 10 and 11 show what happens when we execute the Python script—note the co-existence of Dialog and Xdialog widgets in the screenshots.

## Some other similar utilities

In this section, we are going to touch on some other utilities similar to Dialog and Xdialog. I found `gtkdialog` and `xmessage` preinstalled on Puppy Linux, and Zenity and Whiptail came preinstalled with Ubuntu.

As mentioned earlier, Whiptail provides fewer options and widgets than Dialog. It only provides yes/no, menu, input, message, text, info, checklist, radiolist, gauge and password boxes. Type the following command to see the password widget of Whiptail in action:

```
whiptail --fb --title "Whiptail Password Widget" \
```



Figure 12: Whiptail and Zenity in action

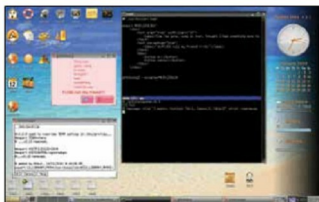


Figure 13: Gtkdialog and Xmessage in action

```
--passwordbox "Enter your password" 20 0
```

Zenity is another utility we can see as a subset of Xdialog. It provides some decent GTK+ widgets and common options in terms of functionality. Type the following commands to see it in action:

```
zenity --info --text"This is an info from Zenity." &
zenity --warning --text"This is a warning from Zenity." &
zenity --question --text"This is a question from Zenity." &
```

Now take a look at Figure 12 to see both Whiptail and Zenity in action.

Coming to Xmessage, it's actually a command to provide a very limited-functionality GUI widget. It only shows a message with user-defined buttons. Run the following command:

```
xmessage -file ~/.bashrc --buttons "Ok:1, Cancel:2, Help:3" -print -nearmouse
```

The `-file` option reads information from a supplied file; and we create various buttons by providing a comma separated list of `title:returcode` to the `-buttons` option. On clicking a button, the corresponding return code is returned. So on the basis of the returned value, we can do further processing.

The functionality of Gtkdialog is very different from all the utilities we have discussed till now. This one creates